Hemant Kumar

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Research Summary

My research lies at the intersection of robot learning, multimodal perception and control theory, with the goal of developing robotic systems that can reason, adapt and act reliably in unstructured real-world environments. To advance this vision, I work across foundation models, world models, and policy learning, connecting vision-language representations, action-conditioned reasoning, and control-theoretic principles to enable safe and generalizable robot behavior.

Education

University of Maryland, College Park Sept 2022 – Present
Ph.D. in Mechanical Engineering Advisor: Dr. Nikhil Chopra
Thesis Direction: Autonomous Disassembly of EVs (Electric Vehicles).
Imitation Learning · Reinforcement Learning · Control Theory · VLAs · Truthworthy Robotics · Computer Vision

Indian Institute of Technology (IIT), Kharagpur Sept 2015 – May 2020
Bachelors (Honors) and Masters of Technology - Aerospace Engineering Advisor: Dr. Chetan S. Mistry

Patents

- **Hemant Kumar (Lead Inventor)**, Kaustubh Joshi, Nikhil Chopra, ***RACE: Real-Time Adaptive Camera Intrinsic Estimation*** (US Patent), Invention of the Year Award-Finalist, University of Maryland (2025).
- Piyush Goenka, **Hemant Kumar**, Kaustubh Joshi, Alexander Beyer, Tianchen Liu, Nikhil Chopra, ***UniScrew: Electronically Actuated Universal Electric Screwdriver Adapter for Any Robot Arm*** (US-Patent).

Research Publications/Submissions

Conferences

- **Hemant Kumar**, Jamin, Sangeethaa Sivakumar, Nikhil Chopra, ***Benchmarking Vision-Language Models for Disassembly Tasks***, The Conference on Artificial Intelligence, ***AAAI 2027***, (under-review).
- **Hemant Kumar**, Nikhil Chopra, Generalizable Adaptive Camera Calibration for Zoom-Adaptive 3D Gaussian Splatting, IEEE/CVF Conference on Computer Vision and Pattern Recognition, **CVPR 2027** (submitting).
- **Hemant Kumar**, Piyush Goenka, Nikhil Chopra, ***DAWN-Visuomotor Diffusion policy for Imitation Learning for Contact Rich Manipulation***, IEEE International Conference on Robotic and Automation, **ICRA** (under-review).
- **Hemant Kumar**, Chetan S. Mistry, ***Implementation of Similarity Principle and Scaling Laws for Low-to-High Speed Mixed Flow Compressor***, American Society of Mechanical Engineers Turbo Expo Boston 2023 *[Link]*.
- **Hemant Kumar**, Chetan S. Mistry, ***Numerical investigations on aerodynamic design criteria for low speed mixed flow compressor***, American Society of Mechanical Engineers (ASME) GTIndia 2021. *[Link]*
- **Hemant Kumar**, Chetan S. Mistry, ***Numerical Performance and Flow Field study of Centrifugal Compressor with Supercritical Carbon dioxide***, American Society of Mechanical Engineers (ASME) GTIndia 2019. *[Link]*.

Journals

- **Hemant Kumar**, Kaustubh Joshi, Nikhil Chopra, ***RACE: Real-Time Adaptive Camera-Intrinsic Estimation via Control Theory***, The IEEE Transactions on Robotics, ***T-RO-2026***, (under-review).
- **Hemant Kumar**, Kaustubh Joshi, Nikhil Chopra, ***From Offline to Online: Adaptive Camera Calibration in Real Time.***, IEEE Robotics and Automation Letters, **RA-L-2026** (under-review).
- **Hemant Kumar**, Nikhil Chopra, ***Online Adaptive Calibration for Pan-Tilt-Zoom Cameras.***, IEEE Robotics and Automation Letters, **RA-L-2026** (under-review).
- **Hemant Kumar***, Piyush Goenka*, Nikhil Chopra, ***Toward Generalizable Robotic Disassembly: A Visuo-Tactile Pipeline for Screw Unfastening***, Journal of Robotics and Autonomous Systems, ***RAS 2026***, *[Link]*.

- **Hemant Kumar**, Chetan S. Mistry, ***Tip Clearance Mechanism in Mixed Flow Compressor***, Journal of Propulsion and Power Research 2023. *[Link]*.

Selected Research Projects

- **DAWN: Hierarchical Diffusion Policies for Contact-Rich Robotic Unfastening**
  - Proposed **DAWN**, a ***hierarchical diffusion based framework*** for autonomous robotic unfastening task, enabling structured learning of long-horizon dexterous manipulation.
  - Collected a **large scale real-world dataset** and evaluated DAWN on **1500+ physical trials** on a **Franka Emika robot**, demonstrating strong generalization across unseen fasteners and operating conditions.
- **Benchmarking Vision-Language Models for Disassembly Tasks**
  - Multimodal benchmark of 68 real-world assembly manuals with 136 assembly/disassembly task graphs to evaluate 20+ foundation models on **entity grounding, procedural reasoning, and disassembly inference**, revealing a consistent gap between local dependency reasoning and globally coherent planning.
- **Structure-Aware Learning for Procedural Reasoning** *(Ongoing Research)*
  - Developing structure-aware methods to improve procedural reasoning in multimodal foundation models, particularly long-horizon dependencies and parallel execution.
- **Adaptive Robot Policies under Changing Physical Dynamics**
  - Developing parameter-efficient online adaptation of manipulation policies using adaptive LoRA to handle changes in physical dynamics and task conditions.
- **Adaptive Control Wrapper for Foundation Robotics Models**
  - Developing a ***control-theoretic wrapper over foundation robotics models***, single-policy networks, and VLA models to enable real-time adaptation under uncertainty in contact-rich manipulation tasks.
- ***A Unified Framework for Real-time Adaptive Camera Calibration for Non-linear Wide Angle Camera Models***
  - A unified adaptive calibration framework for nonlinear wide-angle camera models, enabling **real-time self-calibration with convergence guarantees** for reliable 3D reconstruction, mapping, and localization.

Experience

Research Associate | **Turbomachines Research Lab** | ***IIT Kharagpur*** Aug 2020 -May 2022
Supervisor: Prof. Chetan S. Mistry

- Developed physics-based modeling and CFD simulation pipelines (ANSYS) for compressor design and stability analysis; co-authored peer-reviewed publications.

Research Intern | **Turbomachines Research Lab** | ***IIT Kharagpur*** May 2019 - July 2019
Supervisor: Prof. Chetan S. Mistry

- Designed and evaluated a subsonic mixed-flow compressor using analytical modeling and CFD-based performance optimization.

Research Intern | **Turbomachines Research Lab** | ***IIT Kharagpur*** May 2018 - July 2018
Supervisor: Prof. Chetan S. Mistry

- Built theoretical and numerical models for supercritical CO_2 centrifugal compressors and validated simulations against experimental data.

Research Intern | **DRDO** | ***Chandigarh, India*** May 2017 - July 2017
Supervisor: Scientist 'E' Munesh Kumar Patle

- Modeled and simulated PDE engine dynamics using numerical methods and 3D CFD simulations

Skills

Languages: Python, MATLAB, Familiar with C & C++.

Technologies: Git, Docker, LATEX, MAVLink, Arduino.

Robot Platforms: Universal Robots (UR5e and UR3e), Franka Emika (FR3), Neuromeka IndyRP2.

Robot Software & Simulation: ROS, ROS-2, Gazebo, NVIDIA Isaac Sim.

Vision/AR/VR Systems: Intel Realsense, Quest 3, Real-time object detection & Navigation system integration.

Machine Learning Frameworks: Diffusion-based Imitation Learning, Offline RL, Robot Foundation Models, VLAs.

Past Research and Projects

- **Advanced Machine Learning: Theory and Applications**, University of Maryland (Feb–June 2023) - Developed deep neural and reinforcement learning algorithms for autonomous course navigation.
- **Equilibrium Programming**, University of Maryland (Feb–June 2023) - Implemented a bi-level optimization framework using game-theoretic principles for energy market.
- **AIAA Design Competition 2019** - Ranked among the **Top 8 globally** for hybrid-electric UAV engine design, organized by the American Institute of Aeronautics and Astronautics.

Academic Service

Reviewer : IROS, ICRA, RA-L, ICRA'25, CPHS'26, Robot. Comput. -Integr. Manuf, ASME.
Graduate Teaching Assistant :ENME202: Computing Fundamentals for Eng3, UMD, College Park. *Fall 2026*.
Graduate Teaching Assistant : ENME202: Computing Fundamentals for Eng, UMD, College Park. *Spring 2026*.
Graduate Teaching Assistant : ENES221: Dynamics, UMD, College Park. *Spring 2025*.
Graduate Teaching Assistant : ENES102: Mechanics-I, UMD, College Park. *Fall'22, Spring'23, Fall'23*
Graduate Teaching Assistant : AE39003: Structure Lab-II, IIT, Kharagpur. *Fall 2020*.

Leadership & Honors

- **Future Faculty Program Travel Award**, University of Maryland. *2026*
- **Invention of the Year Award, Finalist**, UMD'25, for RACE: Real-time Adaptive Camera Estimation. *2025*
- **Dean's Fellowship**, University of Maryland, College Park. *2023*
- Conferred the prestigious **Institute Blue Award** for outstanding contribution to Basketball, IIT Kharagpur. *2020*
- Recipient of the **AICTE Postgraduate Scholarship**, Government of India, for academic excellence. *2019-2020*
- **Captain**, IIT Kharagpur Men's Basketball Team - **Silver Medal**, 54th Inter IIT Sports Meet. *2019–20*
- **Group Leader**, National Service Scheme (NSS) - led community service initiatives. *2016–17*